Musterlösung zu Geometrie & Projektion

a) $\vec{AB} = \begin{pmatrix}3\\0\\1\end{pmatrix} - \begin{pmatrix}2\\-\sqrt{2}\\2\end{pmatrix} = \begin{pmatrix}1\\\sqrt{2}\\-1\end{pmatrix}$

$\vec{AC} = \begin{pmatrix}3\\-\sqrt{2}\\1\end{pmatrix} - \begin{pmatrix}2\\-\sqrt{2}\\2\end{pmatrix} = \begin{pmatrix}1\\0\\-1\end{pmatrix}$

b) $\vec{AB} \cdot \vec{AC} = \begin{pmatrix}1\\\sqrt{2}\\-1\end{pmatrix} \cdot \begin{pmatrix}1\\0\\-1\end{pmatrix} = 1 \cdot 1 + \sqrt{2} \cdot 0 + (-1)(-1) = \underline{2}$

$|\vec{AB}| = \sqrt{1+2+1} = \sqrt{4} = 2$

$|\vec{AC}| = \sqrt{1+0+1} = \sqrt{2}$

$\cos(\alpha) = \frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}||\vec{AC}|} = \frac{2}{2 \cdot \sqrt{2}} = 1/\sqrt{2}$

$\underline{\alpha = \arccos\left(1/\sqrt{2}\right) \approx 45°}$

c) $A = 1/2 \underbrace{|\vec{AB} \times \vec{AC}|}_{\text{Fläche d. Parallelogramms}} = 1/2 \left| \begin{pmatrix}1\\\sqrt{2}\\-1\end{pmatrix} \times \begin{pmatrix}1\\0\\-1\end{pmatrix} \right| = 1/2 \left| \begin{pmatrix}-\sqrt{2}\\0\\-\sqrt{2}\end{pmatrix} \right|$

$A = 1/2 \sqrt{2+0+2} = 2/2 \quad \Rightarrow \quad \underline{A = 1}$

d) $\vec{AC}_{\parallel} = (\vec{AC} \cdot \vec{e}_{AB}) \cdot \vec{e}_{AB} = (\vec{AC} \cdot \frac{\vec{AB}}{|\vec{AB}|}) \cdot \frac{\vec{AB}}{|\vec{AB}|}$

$= \frac{1}{2^2} \underbrace{\left[\begin{pmatrix}1\\0\\-1\end{pmatrix} \begin{pmatrix}1\\\sqrt{2}\\-1\end{pmatrix} \right]}_{=2} \cdot \begin{pmatrix}1\\\sqrt{2}\\-1\end{pmatrix} = 1/2 \begin{pmatrix}1\\\sqrt{2}\\-1\end{pmatrix} = \underline{\begin{pmatrix}1/2\\\sqrt{2}/2\\-1/2\end{pmatrix}}$

$\vec{AC}_{\perp} = \vec{AC} - \vec{AC}_{\parallel} = \begin{pmatrix}1\\0\\-1\end{pmatrix} - \begin{pmatrix}1/2\\\sqrt{2}/2\\-1/2\end{pmatrix} = \underline{\begin{pmatrix}1/2\\-\sqrt{2}/2\\-1/2\end{pmatrix}}$

e) $A = 1/2 \cdot |\vec{AB}| \cdot |\vec{AC}_{\perp}| = 1/2 \cdot 2 \cdot \sqrt{(1/2)^2 + (\sqrt{2}/2)^2 + (-1/2)^2} = \frac{1}{2} \cdot 2 \cdot 1$

$\underline{A = 1}$